Risk Alert Classifier

Part A: Conceptual Understanding (Theory)

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1. What is Logistic Regression and why is it suitable for classification?

Logistic Regression is a supervised model that passes a linear combination of the features through the sigmoid function, σ(z) = 1 / (1 + e^(−z)), to output the probability that a sample belongs to the positive class (here, High Risk). A threshold (default 0.5) turns this probability into a class label.

It suits classification because its output is bounded between 0 and 1, it is trained by maximizing likelihood (log-loss), and its coefficients are interpretable as log-odds. This makes it a fast, transparent baseline for risk scoring, and the threshold can be tuned to business needs.

2. Explain classification performance metrics and why accuracy alone is insufficient.

Classification metrics (confusion matrix, accuracy, precision, recall, F1-score, ROC-AUC) summarize how well predicted labels match actual labels. Each one captures a different kind of correctness or error.

Accuracy is the share of all predictions that are correct, so it hides the type of error made. If only 5% of customers are high risk, a model that labels everyone Low Risk scores 95% accuracy yet catches no risky customer. Recall, precision, F1 and AUC are needed to judge minority-class performance.

3. Define Type-I Error and Type-II Error in the context of risk prediction.

Type-I Error (False Positive): a genuinely low-risk customer is flagged as high risk. The cost is customer inconvenience, wrongly blocked or declined transactions, extra manual review and possible loss of a good customer.

Type-II Error (False Negative): a genuinely high-risk customer is predicted as low risk and missed. The cost is undetected defaults or fraud and direct financial loss. For an early-warning system this is usually the more expensive error, so the model should minimize false negatives.

4. Explain Precision, Recall, F1-Score, TPR, and FPR.

Precision = TP / (TP + FP): of the customers flagged as high risk, the fraction that truly are. Recall = TP / (TP + FN): of all truly high-risk customers, the fraction the model catches. F1-Score = 2·P·R / (P + R): the harmonic mean of precision and recall, which balances the two.

TPR (True Positive Rate) is the same as Recall (sensitivity). FPR = FP / (FP + TN): the fraction of low-risk customers wrongly flagged as high risk. TPR and FPR are the axes of the ROC curve.

5. What is AUC-ROC and how does it help in evaluating classifiers?

The ROC curve plots TPR against FPR at every possible classification threshold. AUC (Area Under the Curve) condenses it into one number from 0 to 1. 0.5 means random guessing and 1.0 means perfect separation. It can be read as the probability that the model ranks a random high-risk customer above a random low-risk one.

AUC is threshold-independent, so it lets us compare models fairly before choosing an operating threshold, and it is less distorted by class imbalance than accuracy. With severe imbalance, the precision-recall curve is a useful complement.

6. Why does imbalanced data create problems in classification models?

When one class heavily outnumbers the other, the model sees few minority examples, so the loss is dominated by the majority class. It learns to favor the majority, giving high accuracy but low recall for the high-risk class, which is the class we care about most.

Decision boundaries become biased toward the majority class and standard metrics become misleading. Remedies include resampling (under-sampling, over-sampling, SMOTE, ADASYN), class weights, threshold tuning and evaluating with recall, F1 and AUC instead of accuracy.